Determining Planck and Boltzmann constants

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In this experiment, we measured two physical constants, namely the Planck constant h and the Boltzmann constant k_B. For calculating the Planck constant, we measured five characteristics of a photo diode that is being shot with light of different wavelengths. Reading the stopping potential of each series of measurement we determined the Planck constant $h = (6.51 \pm 0.15) \cdot 10^{-34}\,\mathrm{J\,s}$. By measuring the diffusion current characteristics of a transistor at different temperatures, we were able to calculate the Boltzmann constant to $k_B = (1.403 \pm 0.038) \cdot 10^{-23}\,\mathrm{J/K}$.

1 Introduction

For the description of nature many natural constants are necessary. This paper deals with the identification of two of them, the Planck and the Boltzmann constant. In quantum mechanics the Planck quantum is essential to describe harmonic oscillating processes of systems, which are able to do so. Moreover, the Boltzmann constant is crucial in statistical mechanics and thermodynamics, because it connects entropy with the count of micro states in a state space.

2 Basics

The Planck constant h is defined with the Planck-Einstein relation

$$E = h\nu \tag{1}$$

where E is the energy of a quantum and ν the oscillation frequency of it [1]. There are different types of quantums like phonons or photons and all of them can only have integer multiples of E. With the photo effect one can measure h indirectly. If monochromatic light with corresponding energy E hits metal, the metal can be ionised and as a consequence electrons of the atom, which have exactly the same binding energy are released. With equation

$$\nu = \frac{c_0}{\lambda} \tag{2}$$

one can see that the intensity of the light is not responsible for this process, but rather the wave length λ and the speed of light in vacuum $c_0 = 299\,792\,458\,\mathrm{m/s}$. In an electric field electrons have a potential energy, so putting an anode and a cathode around the metal, leads to

$$E = eU_c = h\nu - W_e \tag{3}$$

where W_e is the work function, which is the work needed to ionise an electron, e is the elementary charge and U_c is the cut off potential. In this case U_c was used as a stopping potential, so one can determine the kinetic energy of a detected electron.

For determining the Boltzmann constant k_B, one can use a diode in the direction, where a current I is blocked. This corresponds to a p-n junction, which has an area with electron abundance and an area with defect abundance (a positive charge is called defect). Between those areas a sector forms without any charges, followed by a potential difference U_A. If a positive or negative charge has an energy E, which accomplishes

$$E > eU_A \tag{4}$$

it can pass the sector and causes a current I. The abundance distribution $N(E)$ for those particles is the Boltzmann distribution and integrating over all energies E which fulfills the inequality 4, leads to the diffusion current I_D

$$I_D(U - A) = b\ ek_BT\ exp\left(\frac{eU_A}{k_BT}\right) \tag{5}$$

where b is a constant and T the temperature [2]. Without an external potential U, the diffusion current nullifies the intrinsic conduction current I_0, leading to $I_D(U_A) = |I_0|$. With an external potential orientated in the direction where a current can pass the diode, I_D only have to overcome the potential energy $E = e(U_A - U)$, which yields with eq. 5 the Shockley's equation for a total current I [2]

$$I(U) = |I_0|\left(exp\left(\frac{eU}{k_BT}\right) - 1\right) \tag{6}$$

3 Execution

3.1 Planck constant

A mercury lamp is used to shine light through a lens collimating the light on the phototube. Between the

lens and the phototube, different monochromatic filters are placed. For each of these filters, a series of measurements is conducted by varying the stopping potential from $U = -3\,\mathrm{V}$ to $U = 0\,\mathrm{V}$ or until the photocurrent exceeds $I_{ph} = 1\,\mathrm{nA}$. These measurements resulted in the values plotted in fig. 1. One can see two phenomena here. The first one is that the current flowing before the cut off potential is reached gets smaller with increasing wavelength. This is due to the lower wavelength photons, which have higher energy. The higher energy of the photon makes it possible to ionise an atom in the anode, which creates an additional current. For further discussion, this can be ignored.

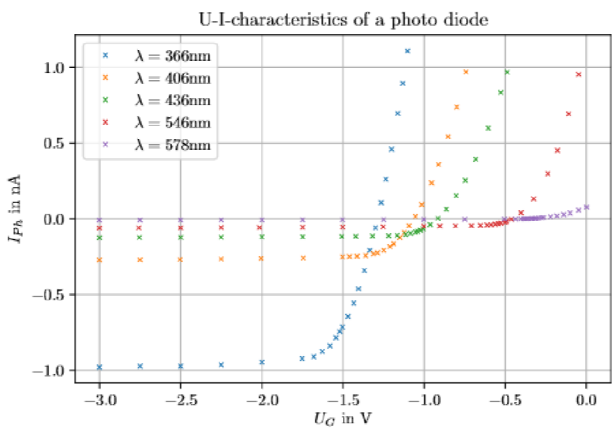

Fig. 1: The five characteristics of a photo diode at different wavelengths. U is the negative potential applied between anode and cathode. One can see the cut off point increasing with increasing wavelength.

Second, the cut off potential is higher when the wavelength is higher. This relation is explained by the EINSTEIN equation (eq. 3), that states a linear relation between $\nu = c/\lambda$ and the stopping potential U, with the slope m and the intercept y_0

$$m = -\frac{h}{e}, \quad y_0 = \frac{W_e}{e} \tag{7}$$

At first, the cut off potential needs to be examined. For that, the linear part of the characteristic curves is fitted linear, with a second parallel line resembling the maximum error of the first line. A polynomial fit is used to approximate the value of the cut off potential, while the right-most point of the linear fit is used for determining the error of the reading. One example of determining the cut off potential is shown in fig. 2.

Second, the values calculated like that are plotted in fig. 3 against the frequency ν. The linear regression leads to the following results with eq. 7 for h and W_e

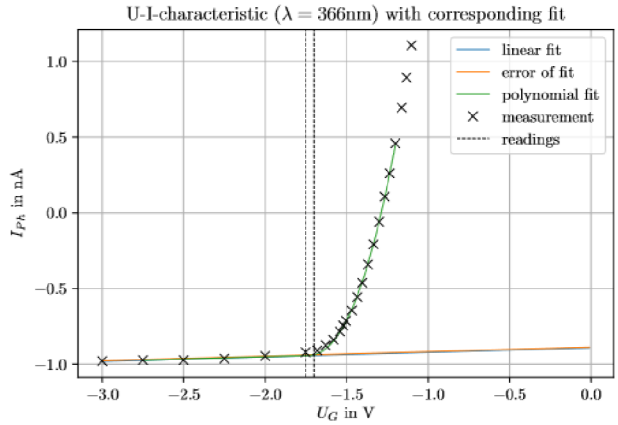

Fig. 2: Example of a characteristic at $\lambda = 366\,\mathrm{nm}$, showing the way we determined the cut off potential. After selecting a cut off point and doing a linear fit with the points left to it, we calculated a parallel line to this fit representing the fit error. Using a polynomial fit to interpolate the points right from the cut off point, we read the intersection of the error line with this fit to determine the cut off potential.

and has a correlation coefficient of $R = 0.999\,203\,13$.

$$\begin{aligned} h = -me &= -e \cdot (-4.065 \pm 0.094) \cdot 10^{-15}\,\mathrm{V\,s} \\ &= (6.51 \pm 0.15) \cdot 10^{-34}\,\mathrm{J\,s} \\ W_e &= (1.646 \pm 0.063)\,\mathrm{eV} \end{aligned}$$

The reference value for the planck constant is $h = 6.626\,070 \cdot 10^{-34}\,\mathrm{J\,s}$[1], which fits our measurement in the range of its errors. The work function extrapolated here can be found in CaO, so it is possible that the photo cathode is made from calcium oxide. Further tests on this were not possible in the range of this experiment.

3.2 Measuring the Boltzmann constant

In this section a npn transistor upon a thermostat *RE104* is used to realise the necessary p-n junction, discussed in sec. 2. With the thermostat the temperature inside the transistor is regulated to a constant value, while the measurement is beeing taken. The npn transistor is required to reduce the noise currents I_{Stoer} caused by the materials and the setup, making a more detailed experiment possible. A ground at the p doped sector leads to a turning of I_{Stoer} from the emitter to the base.

Furthermore, the variables in fig. are U_{BE} as an adjustable base emitter potential and I_{Diff} as diffusion current I_D, which equals the collector current I_C. Measuring I_C at the collector for three different temperatures, Shockley's equation (eq. 6) reduces

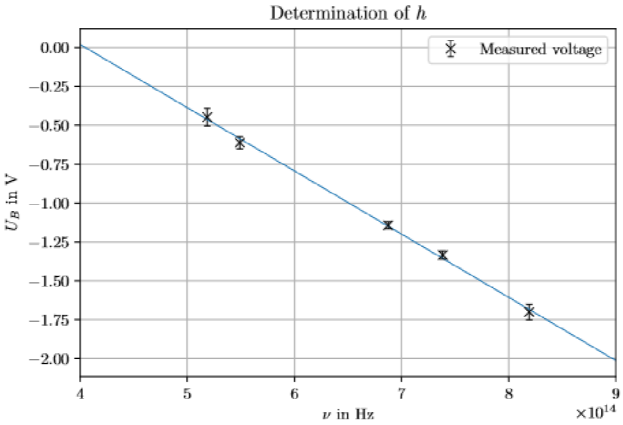

Fig. 3: The cut off potentials were plotted against the frequency of the light. Fitting a straight line and determining its slope we could calculate h.

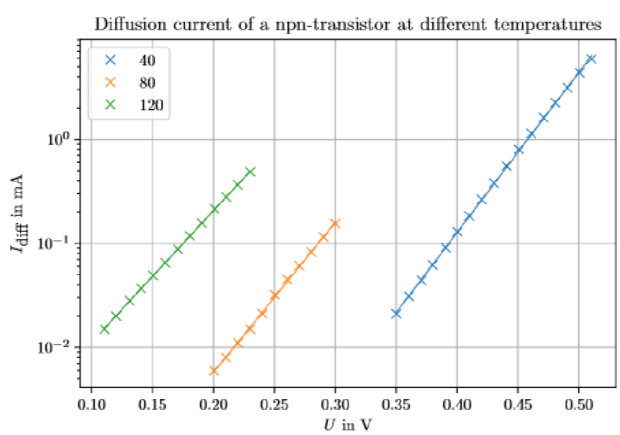

Fig. 4: The three characteristics of a transistor at different temperatures in a linear-logarithmic plot. The slopes of the linear fits were used to determine the Boltzmann constant k_B.

to the first term, allowing a semi-logarithmic plot. The results for different potentials U_{BE} can be seen in fig. 4. Fitting a linear function to the series of measurements, one can determine the Boltzmann constant by

$$k_B = \frac{e}{mT} \ln(e)\,, \tag{8}$$

where ln is the natural logarithm. Determining the slopes and calculating k_B for every series of measurement gives the values listed in tab. 1.

Averaging these values gives the Boltzmann constant

$$k_B = (1.403 \pm 0.038) \cdot 10^{-23}\,\mathrm{J/K}\,.$$

Within the frame of error, the measured value is comparable with references $k_{B,\mathrm{lit}} = 1.380\,65 \cdot 10^{-23}\,\mathrm{J/K}$ [1]. To minimize the deviation further on, taking more measurements or a more sensitive thermometer could be used.

4 Conclusion

In this experiment it has been possible to measure two of the most important physical constants. Measuring the Planck constant using the photoelectrical effect gave $h = (6.51 \pm 0.15) \cdot 10^{-34}\,\mathrm{J\,s}$, while measuring the Boltzmann constant with the diffusion current of a transistor gave $k_B = (1.403 \pm 0.038) \cdot 10^{-23}\,\mathrm{J/K}$. These two measurements match the reference values within their error ranges with a deviation of 1.70 % and 1.61 % respectively. The results can be regarded as a good demonstration of how precise fundamental physical constants can be determined using simple methods.

References

[1] D. Meschede. *Gerthsen Physik*. Springer-Lehrbuch. Springer Berlin Heidelberg, 2015.

[2] University of Wuerzburg. *Plancksches Quantum und Boltzmann-Konstante*, 2019.

temperature	**slope**	k_B
$T = (40.00 \pm 0.10)\,°\mathrm{C}$	$(35.49 \pm 0.16)/\mathrm{V}$	$(1.4417 \pm 0.0066) \cdot 10^{-23}\,\mathrm{J/K}$
$T = (80.00 \pm 0.10)\,°\mathrm{C}$	$(33.24 \pm 0.32)/\mathrm{V}$	$(1.365 \pm 0.013) \cdot 10^{-23}\,\mathrm{J/K}$
$T = (120.00 \pm 0.10)\,°\mathrm{C}$	$(29.070 \pm 0.074)/\mathrm{V}$	$(1.4019 \pm 0.0037) \cdot 10^{-23}\,\mathrm{J/K}$

Tab. 1: The results for the evaluation of the Boltzmann constant.